INCH-POUND

MIL-DTL-50911B
7 January 2011
SUPERCEDING
MIL-P-50911A (AR)
19 March 1983

DETAIL SPECIFICATION

PRIMING ASSEMBLY FOR DEMOLITION CHARGE ASSEMBLY: M183 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

Reactive after 7 January 2011 and may be use for new design and existing design and acquisition

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification prescribes the requirements and identifies the verification methods for the Priming Assembly for Demolition Charge Assembly: M183 Parts For and Loading, Assembling and Packing.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: RDAR-QES-E, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1375

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-17124	-	Cord, Detonating

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

8861109	-	Clip, Cord, Detonating, M1
8861119	-	Booster, Assembly
8861120	-	Cup Booster
8861134	-	Fuse, Detonating
9219074	-	Priming Assembly

(Copies of these drawings are available from U.S. Army ARDEC, ATTN: RDAR-EIS-PE, Picatinny, NJ 07806-5000, or email pica.drawing.request@conus.army.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A36/A36M	-	Standard Specification for Carbon Structural Steel

(Copies of these documents are available from http://www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of the Priming Assembly shall be subjected to first article inspection in accordance with 4.3.

3.2 Conformance inspection. A sample of the Priming Assembly shall be subjected to conformance inspection in accordance with 4.4.

3.3 Priming assembly. The priming assembly shall be in accordance with drawing 9219074.

3.4 Detonating fuse. Each lot of the detonating fuse shall be in accordance with MIL-DTL-17124.

3.5 Weight of charge. The weight of the booster charge shall be in accordance with drawing 8861119.

3.6 Output. Both boosters of one individual priming assembly shall initiate the M112 demolition charge by resulting in cutting through or indenting steel witness plates.

3.7 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.8 Workmanship. Al1 parts and assemblies shall be free of burrs, sharp edges, cracks, dirt, and other foreign matter. The booster cups shall not be cracked, bulged, scratched or deformed. Charges shall be free of breaks, cracks and other defects. Colors shall be uniform. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

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4. VERIFICATION

TABLE I. Requirement/Verification cross reference matrix

Methods of verification		Classes of verification						
1 – Analysis 2 – Demonstration 3 – Examination 4 – Test		A – First Article B – Conformance						
Section 3 Requirements	Description	Verification Method				Verification Class		Section 4 Verification Method
		1	2	3	4	A	B	
3.1	First article			X	X	X		4.3
3.2	Conformance inspection			X	X		X	4.4
3.3	Priming assembly			X		X	X	4.4.2.1 through 4.4.2.4
3.4	Detonating fuse	X	X	X	X	X	X	4.4.2.4
3.5	Weight of charge				X	X	X	4.4.2.3, 4.5.2
3.6	Output				X	X	X	4.5.3
3.7	Ammunition lot number			X		X	X	4.5.4
3.8	Workmanship			X		X	X	4.4.2.1 through 4.4.2.4, 4.5.5

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.3)
b. Conformance inspection (see paragraph 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in 4.5.

4.3 First article. When specified, a sample of the priming assembly and subcomponents shall be subjected to first article inspection in accordance with Table II.

4.3.1 First article quantity. The first article sample shall consist of the following items in sample quantities as indicated in Table II.

4.3.2 First article rejection. If any assembly, component or test specimen fails to comply with requirements, the first article sample shall be rejected (see 6.7).

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TABLE II. First article tests and inspection.

Examination or Test	Drawing Number	Sample Size	Requirement Paragraph	Verification
M1 detonating cord clip	8861109			
Examination of Defects		10	3.3	4.4.2.1
Booster cup	8861120			
Examination of Defects		10	3.3	4.4.2.2
Booster assembly	8861119			
Examination of Defects		10	3.3	4.4.2.3
Weight of Charge		10	3.5	4.5.2
Priming assembly	9219074			
Examination of Defects		10	3.3	4.4.2.4
Output		3	3.6	4.5.3

4.4 Conformance inspection.

4.4.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1916. In addition, each inspection lot of priming assemblies shall be produced with components complying with the following constraints:

a. Detonating fuse from not more than one lot interfix number from one manufacturer.
b. Comp A5 from not more than one lot interfix number from one manufacturer.
c. Cup boosters from not more than one lot interfix number from one manufacturer.
d. Clip cords from not more than one lot interfix number from one manufacturer.

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level VII in addition to 100% inspection shall be used for all characteristics defined as Critical, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics.

b. Inspection. Conformance inspection shall be performed in accordance with section 4.4.2.1 through 4.4.2.5. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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4.4.2.1	M1 detonating cord clip			Drawing Number 8861109 Next Higher Assembly 9219074
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method / Reference
Critical	None defined			
Major	None defined			
Minor				
201	Workmanship	Level II	3.8	Visual

4.4.2.2	Booster cup			Drawing Number 8861120 Next Higher Assembly 8861119
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method / Reference
Critical	None defined			
Major				
101	Cup split or cracked			
Minor				
201	Length	Level II	3.3	Gage
202	Thickness through bottom, min	Level II	3.3	Gage
203	Inner diameter	Level II	3.3	Gage
204	Workmanship	Level II	3.8	Visual

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4.4.2.3	Booster assembly			Drawing Number 8861119 Next Higher Assembly 9219074
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method / Reference
Critical				
1	Booster cups split or cracked, exposed charge, evidence of explosive leakage	100%	3.8	Visual
Major				
101	Charge missing	Level IV	3.3	Gage
102	Outside diameter, max	Level IV	3.3	Gage
103	Depth to charge, min	Level IV	3.3	Gage
104	Weight of charge	1/	3.5	4.5.2
Minor				
201	Marking missing, incorrect, or identifiable	Level II	3.3	Visual
202	Workmanship	Level II	3.8	Visual
Note: 1/ Total of 32 samples.				

4.4.2.4	Priming assembly			Drawing Number 9219074 Next Higher Assembly
Classification	Examination or Test	Conformance Criteria	Requirement Paragraph	Inspection Method Reference
Critical				
1	Booster cups split or cracked, exposed charge, evidence of explosive leakage	100%	3.8	Visual
Major				
101	Length, max	Level IV	3.3	Gage
102	Double crimp missing or incorrect	Level IV	3.3	Visual
103	Booster assemblies (2) improperly assembled or missing	Level IV	3.3	Visual
104	Detonating cord clips (2) improperly assembled, damaged, missing, or not secure to detonating fuse	Level IV	3.3	Visual
105	Output	1/	3.6	4.5.3
Minor				
201	Workmanship	Level II	3.8	Visual
Note: 1/ Total of 3 samples.				

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4.5. Methods of inspection.

4.5.1 Detonating fuse. The detonating fuse lot shall be verified in accordance with MIL-DTL-17124.

4.5.2 Weight of charge. Weigh each of the marked empty booster cups to the nearest 0.1 milligram. Calculate the weights and return the sample booster cups to the regular production lot. After loading of the booster cups, the loaded sample booster cups shall be weighed to the nearest 0.1 milligram. Calculate the weight of the booster charge as follows:

Weight = B - A

Where: A = weight of empty booster cup
B = weight of loaded booster cup

4.5.3 Output. A sample of three (3) M112 demolition charges shall be randomly selected from each accepted M112 lot after the charges have been packaged in charge containers. A total of three (3) priming assemblies shall be used to test six (6) booster assemblies. The M112 demolition charge assembly shall be tested against a steel witness plate to verify effectiveness of the priming assembly.

4.5.3.1 Priming assembly setup. The priming assembly shall be stretched into a straight line with the clip cord placed equidistant between the two booster assemblies. A blasting cap capable of initiating the detonating fuse shall be attached and secured to the clip cord. For each priming assembly there are two (2) booster assemblies on each end that shall be tested. Two (2) individual witness plates shall be used for each of the two (2) booster assemblies. Each booster assembly shall be primed and set up in accordance with section 4.5.3.2.

4.5.3.2 Demolition charge setup. The M112 demolition charge shall be cut approximately in half. A steel witness plate of ASTM A36 steel with dimensions of 1.0 x 10.0 x 10.0 inches (nominal) shall be used to verify M112 detonation. The charge shall be applied to the steel witness plate and primed at the center using the booster assembly end of the priming assembly (part no. 9219074). The primed M112 block shall be secured with tape to ensure intimate contact between the charge and the witness plate (see Figure 1. Test Set Up). The detonating fuse shall be secured with tape to the witness plate. The blasting cap shall be initiated and both booster assemblies shall completely initiate each ½ block M112 demolition charge.

4.5.3.2.1 Rejection. Failure of only one of the ½ block M112 demolition charge to cut through or indent the steel witness plate shall result in lot rejection of the priming assembly.

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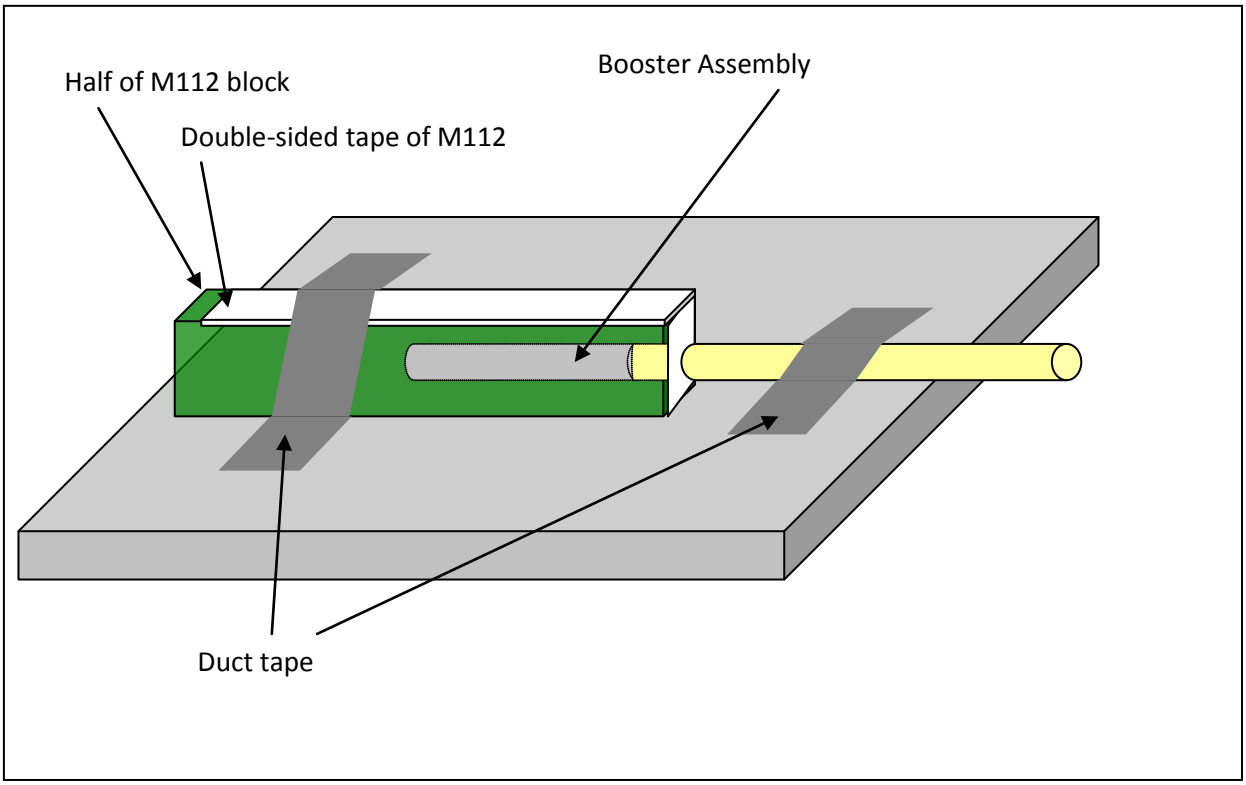


Figure 1. Test Setup.

4.5.4 Ammunition lot numbering. Visually verify that an ammunition lot number has been applied to each lot as described in MIL-STD-1168.

4.5.5 Workmanship. Visually verify that all parts and assemblies meet requirements of 3.8.

5. PACKAGING

5.1 Packaging instruction. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory)

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6.1 Intended use. The Priming Assembly is used to initiate all past and current fielded and developmental demolition items such as explosives, shaped charges, and cratering charges.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Requirement for first article samples.
c. Requirement for conformance inspection.
d. Requirement for Cord, Detonating.
e. Requirements for Ammunition Lot Numbering.
f. Requirement for acceptance inspection equipment (AIE) designs (see 6.3).
g. Certificate of Conformance for each lot or shipment of product.
h. Packaging requirements (see 5.1).

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, U.S. Army ARDEC, ATTN: RDAR-QEM-G, Picatinny Arsenal, NJ 07806-5000. Request letter of submittal should state the contractor, contract number, specification number, item nomenclature and classification of defects or tests. This address should be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Submission of test data. In addition to the normal distribution of records, when the Priming Assembly is procured by the US ARDEC, one (1) copy of the ammunition data card for each lot should be forwarded to: Commander, ARDEC, ATTN: RDAR-QEM-G, Picatinny Arsenal, NJ 07806-5001.

6.5 Performance oriented packaging. Prior to shipment, the contractor should ensure either that the container has been tested by either a U.S. Government approved Performance Oriented Packaging (POP) test laboratory or by the government itself in compliance with POP requirements in accordance with Title 49, Code of Federal Regulations. Test should be to a weight at least 10% greater than the actual gross weight or the certification weight specified on the packaging drawings, (whichever is higher) which should be marked on the tested container. POP marking, however, should not be applied to the container until verified by the U.S. Government as follows:

a. The POP test report should be prepared by the POP test laboratory in accordance with DI-PACK-81059 (Data Item Description) following the test.
b. The report should be kept on file by the contractor and should also be submitted to the U.S. Army (ARDEC Packaging Div) for verification as directed in the contract data requirements list (CDRL) DD Form 1423.
c. The US Army will review the report, prepare an endorsement letter, and submit the package to the Defense Logistics Agency (DLA).

For multi-year contracts, the contractor should re-perform POP testing at a certified POP test laboratory if:

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a. The initial POP test report expires before the end of the contract (single pack POP certification is valid for a period of 1 year; combination pack POP certification is valid for a period of 2 years) or

b. There is a change in container manufacturer, design, or manufacturing processes of the exterior shipping container.

No re-test is needed if all packaging is purchased while under an unexpired POP test.

EXCEPTION TO POP MARKINGS: If the container is manufactured outside the USA, the contractor should not apply the UN POP certification mark provided in this contract (if applicable). The contractor/container manufacturer (outside the USA) is responsible to perform the UN POP certification tests and apply the marking authorized by the Transportation Competent Authority of the country of manufacture. The contractor should also provide a copy of the POP Test Report to the U.S. Army (ARDEC Packaging Div) for concurrence prior to shipment.

6.6 Packaging, marking, and shipping. The following instructions should be used for packing, marking, and shipping of the priming assembly.

6.6.1 Preservation - packaging.

a. Level A. Place 100 Cup, Booster, 8861120, in a box conforming to Variety 2, Process II, of spec PPP-B-566.

b. Level B. Level B should be the same as Level A.

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c. Level C. Cup, Booster should be packaged in accordance with ASTM D3951.

6.6.2 Intermediate pack.

a. Level A. Two hundred boxes (20000 Cup, Booster) should be packed into a fiberboard box conforming to Grade V3c, Class WWVR, Style RSC of ASTM-D5118 and closed per ASTM-D1974.

b. Level B. Level B should be the same as Level A.

c. Level C. Level C, intermediate pack, should be in accordance with ASTM D3951.

6.6.3 Packing. Priming Assemblies should be packed in a manner which will insure arrival at destination in satisfactory condition. The priming assemblies should be wrapped and cushioned to the extent necessary to provide protection from hazards of contamination and physical damage encountered in shipping, handling and storage. In no event should the priming assembly be coiled to form a kink in the detonating fuse or to form a coil less than a 2 inch radius. The shipping container should be in accordance with the rules and regulations of the Code of Federal Regulations, Title 49, Parts 190-199.

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a. Level A. Three intermediate, Level A, containers of Cup, Booster should be over-packed in a wood box conforming to MIL-DTL-2427. Close and strap box in accordance with MIL-DTL-2427.

b. Level B. Level B should be the same as Level A.

c. Level C. Cup, Booster should be packed in accordance with ASTM D3951.

6.6.4 Marking. Marking of Priming Assemblies should be in accordance with MIL-STD-129, and Code of Federal Regulations, Title 49, Parts 100-199, and should include, but not be limited to, the following:

a. Levels A & B. Marking should be in accordance with MIL-STD-129.

b. Level C. Marking of Cup, Booster, should be in accordance with ASTM D3951.

	SIZE (NOM)
NSN and DODIC	1 inch
LOT NUMBER	1 inch
DOT MARKING - DETONATING PRIMERS-HANDLE CAREFULLY	1 inch

6.6.5 Shipping. When components from more than one lot are shipped at one time, each lot should be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6.7 First article rejection. The procuring agency reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

6.8 Critical characteristic. Inadvertent initiation of the booster cups can lead to injury and death. The booster cup being split or cracked, explosive charge being exposed or there being evidence of explosive leakage is a critical characteristic of the Priming Assembly. The explosives contained within the booster cups should remain within the cups.

6.9 Subject term (key word) listing.

Det Cord
Composition C4
C4 block

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Army – AR
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Preparing activity:
Army – AR
(Project: 1375-2009-007)

Review activities:
Navy – NP
Air Force – 70, 99

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